Entropy-Complexity Based Clustering of ARMA Simulated Time Series Models

M. H. M. R. S. Dilhani[1]
Alejandro C. Frery[1]
Andrea A. Rey[2]

[1]School of Mathematics and Statistics, Victoria University of Wellington, New Zealand
[2]LIDEC, Universidad Nacional de Hurlingham (UNAHUR), Argentina

09 December 2025

Motivation and Background

- Time series clustering is challenging because each series can have many possible structures, and random fluctuations make it even more complex.
- Ordinal patterns capture the order relations among consecutive data points.
- **Bandt & Pompe (2002)** [BP02]: introduced **Ordinal Patterns(OP)** and **Permutation Entropy(PE)**(H).
- **Limitation:** PE quantifies disorder but cannot distinguish stochastic from deterministic chaotic dynamics.
- **Extension:** Statistical Complexity (C).
- We can then map a time series into a point in $H \times C$ plane.
- **Goal:** Differentiate AR, MA, and ARMA models

From Time Series to Ordinal Patterns: An Example

- **Example:** Mean monthly humidity in Wellington.
- Original time series:

$$77.3,\ 81.0,\ 82.4,\ 81.7,\ 83.6,\ 85.6,\ \ldots$$

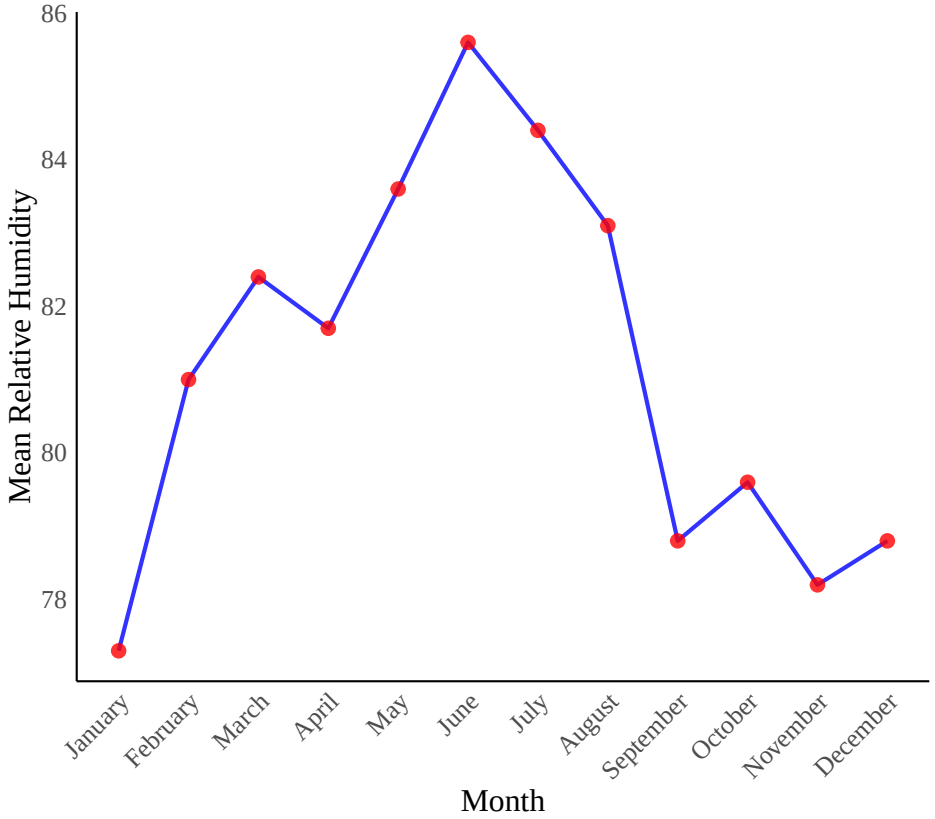
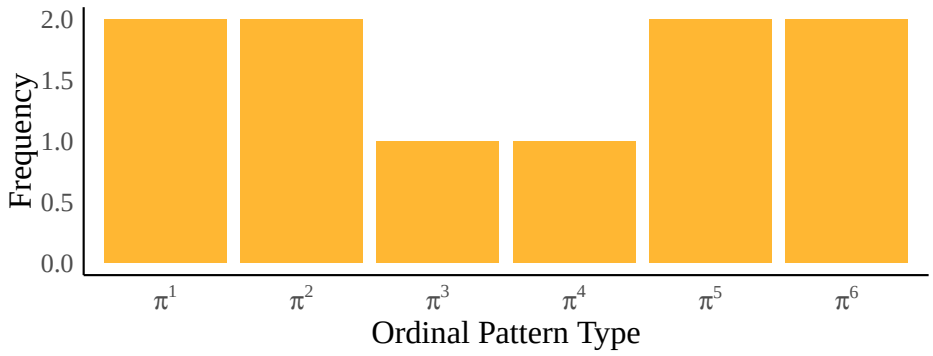


Figure: Mean monthly humidity in Wellington

Transforming into Ordinal Patterns

- For $D = 3$, where D is called as embedding dimension.
- The first window (77.3, 81.0, 82.4) $\rightarrow$ Pattern: (1,2,3) $\rightarrow \pi^1$.
- Next window (81.0, 82.4, 81.7) $\rightarrow$ Pattern: (1,3,2) $\rightarrow \pi^2$.
- Continue for all overlapping windows ($n - D + 1$ total), where n represents the series length.
- Count the frequency of each ordinal pattern.
- There are $3! = 6$ possible patterns: (1,2,3), (1,3,2), (2,1,3), (2,3,1), (3,1,2), (3,2,1).
- The histogram shows the proportion $(p(\pi^\ell))$ of each pattern in the time series.
- Each window represents an ordinal pattern describing local order relations.



Definition: Permutation Entropy (H)

$$H(\mathbf{p}) = -\frac{1}{\log k}\sum_{\ell=1}^{k} p(\pi^{\ell})\log p(\pi^{\ell}),$$

where $k = D!$ and $p(\pi^{\ell})$ is the proportion of pattern π^{ℓ}.

Definition: Normalized Disequilibrium

Given a histogram $\mathbf{p}$ and the reference uniform distribution $\mathbf{u} = (1/k, \ldots, 1/k)$, the normalized disequilibrium is calculated as:

$$Q = \frac{Q'}{\max(Q')},$$

where $Q'(\mathbf{p}, \mathbf{u}) = \sum_{\ell=1}^{k}\left(p_{\ell}\log\frac{p_{\ell}}{u_{\ell}} + u_{\ell}\log\frac{u_{\ell}}{p_{\ell}}\right)$ and $\max(Q')$ is the highest possible value for the given k.

Definition: Statistical Complexity (C)

$$C = HQ$$

Conceptual Framework

- Transform time series $\{x_t\}$ into ordinal patterns.
- Compute:
  - Permutation Entropy
  - Statistical Complexity
  - Confidence Intervals for PE and C
- Plot in the Entropy–Complexity ($H \times C$) plane.
- Visually analyze clusters to discriminate models.

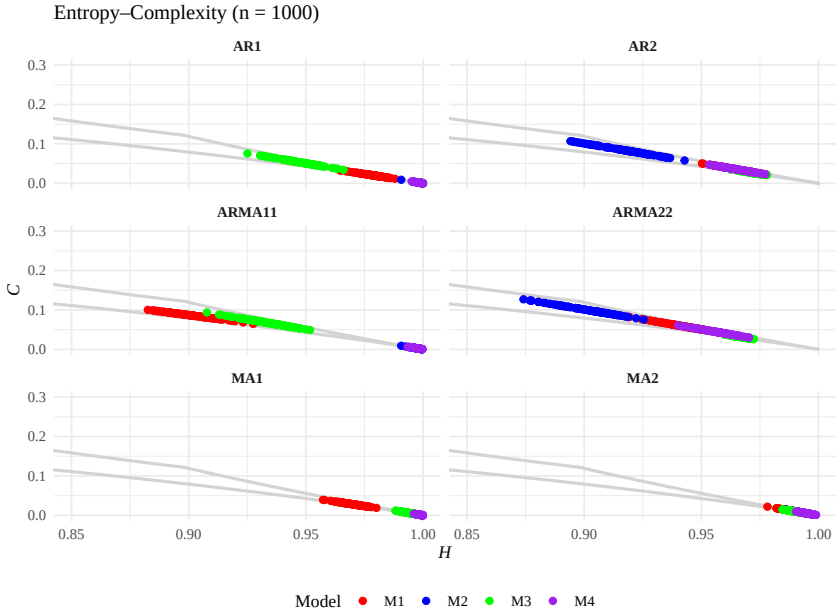


Figure: Entropy–Complexity plane for Sample size $n = 1000$.

Experimental Design

- Models studied:
  - AR(1), AR(2), MA(1), MA(2), ARMA(1,1), ARMA(2,2)
- Each model simulated with 100 independent replications.
- Parameter sets chosen to ensure stationarity and invertibility.
- Series lengths: $n = 1000, n = 5000$.

Model Parameter Structure

Type	Model	α_1	α_2	β_1	β_2
AR(1)	$\mathrm{AR}(1)_{\mathrm{M}_1}$	0.8			
	$\mathrm{AR}(1)_{\mathrm{M}_2}$	0.1			
MA(1)	$\mathrm{MA}(1)_{\mathrm{M}_1}$			0.8	
	$\mathrm{MA}(1)_{\mathrm{M}_2}$			0.1	
ARMA(1,1)	$\mathrm{ARMA}(1,1)_{\mathrm{M}_1}$	0.8		0.8	
	$\mathrm{ARMA}(1,1)_{\mathrm{M}_2}$	0.1		0.1	

Table: Example of parameter sets ensuring diversity.

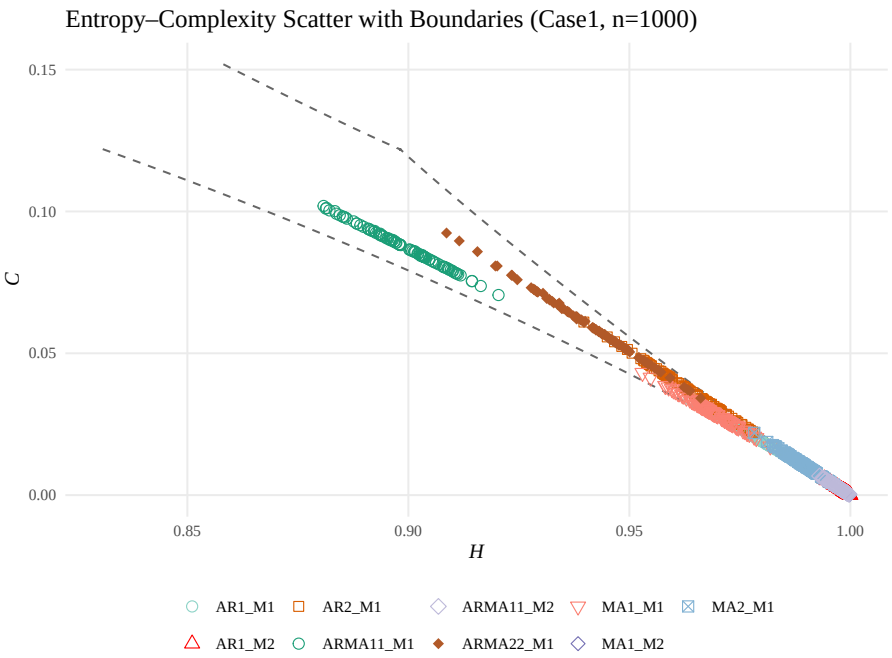
Feature Extraction and Central Points

- For each realization, compute:
  - Entropy (H) — randomness
  - Complexity (C) — structure
  - Confidence Interval(CI) for H and C
- For each model, identify the **central point** (H^*, C^*) using PCA projection.
- Central points correspond to the most representative time series of each model [CQOR+22].

Case Study Design

- **Case 1–3:** Positive, negative, and mixed-sign coefficients.
- Two sample sizes: $n = 1000, n = 5000$.
- Each case analyzed using:
  - The density of points in the $H \times C$
  - Scatter plots with confidence intervals for H and C
  - Visual model discrimination

Case 1 — Positive Coefficients



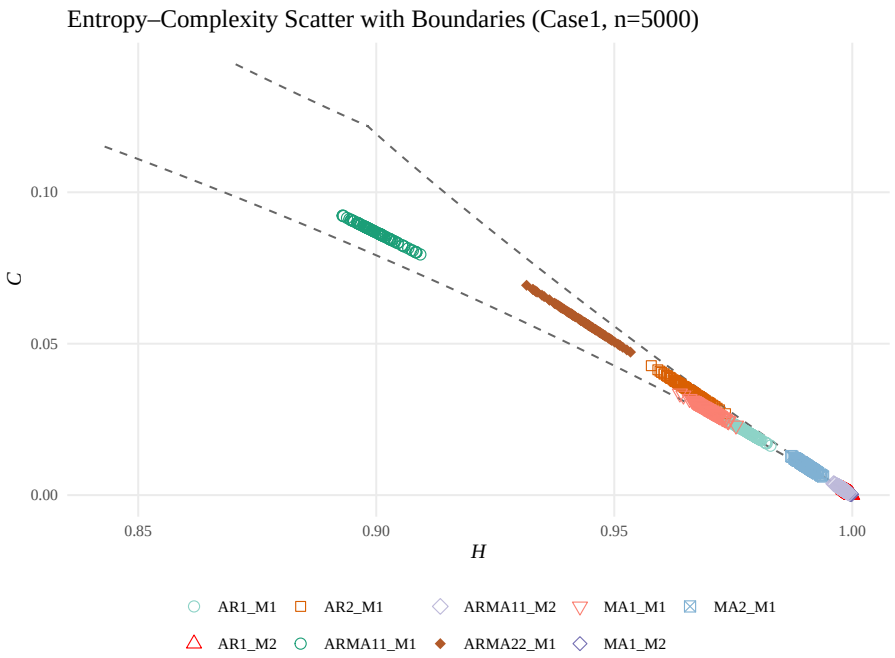
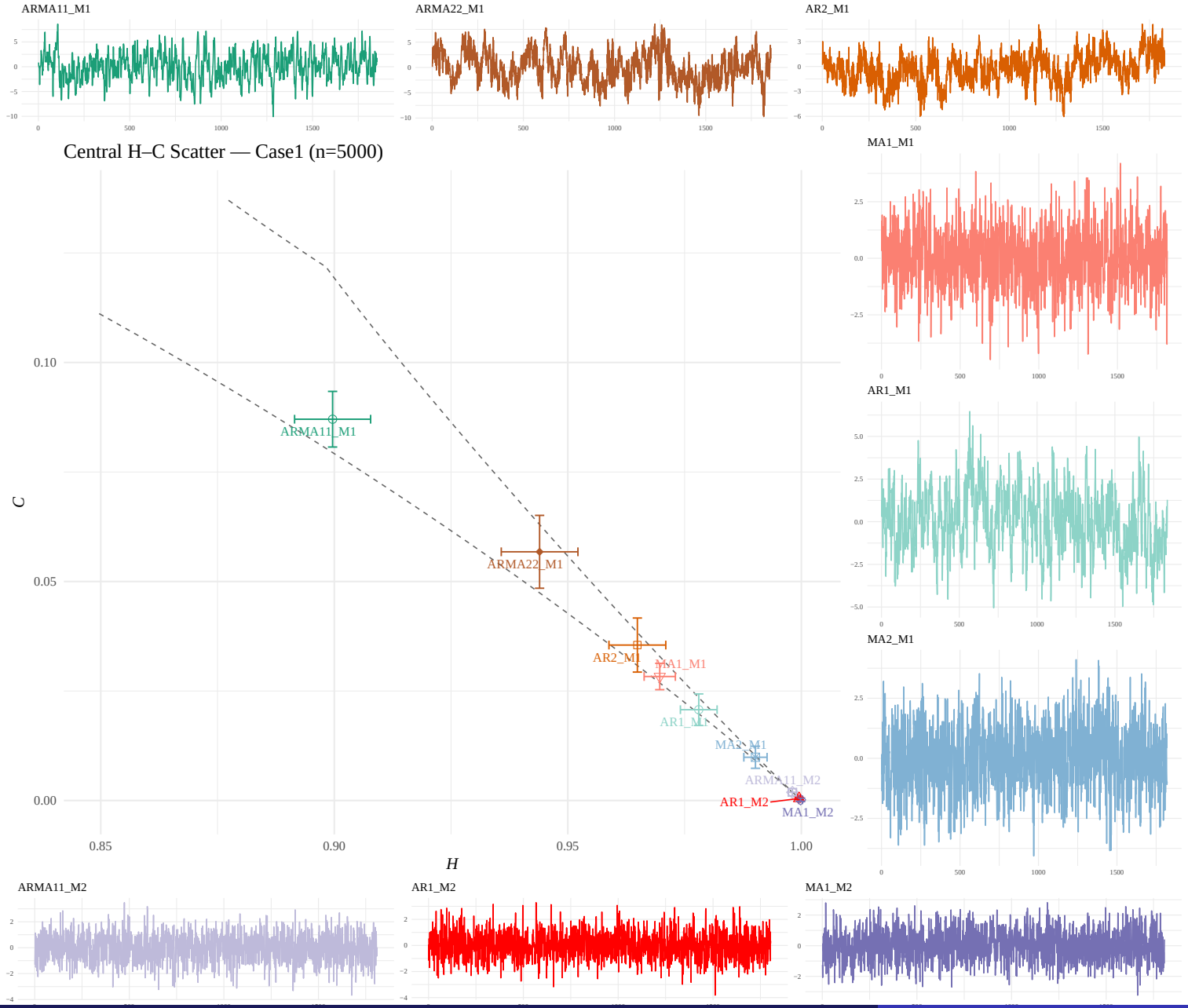
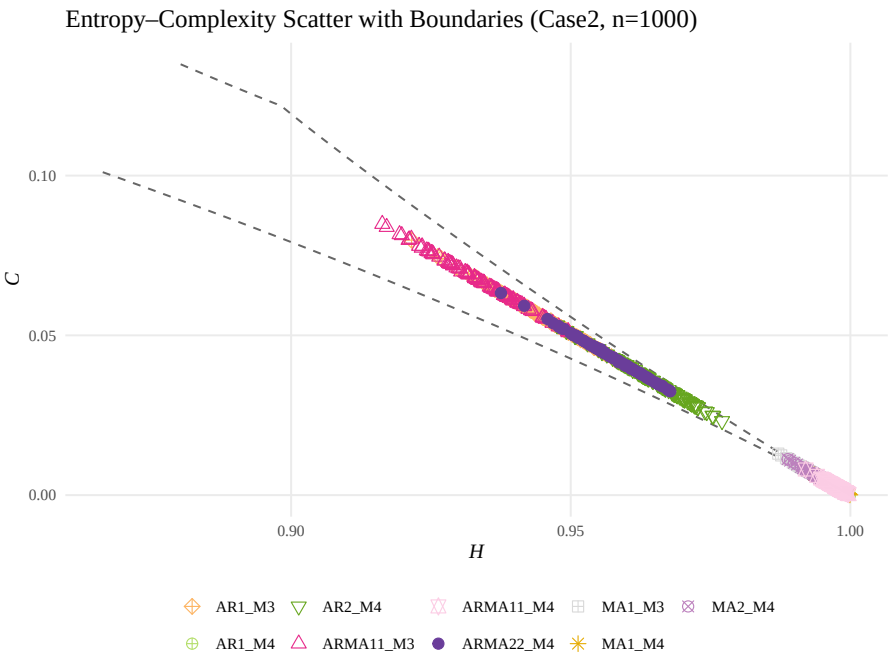


Figure: Entropy–complexity scatter plot ($n = 1000$ and $n = 5000$) for positive model coefficients.

Case 1 — Positive Coefficients cont. . .



Case 2 — Negative Coefficients



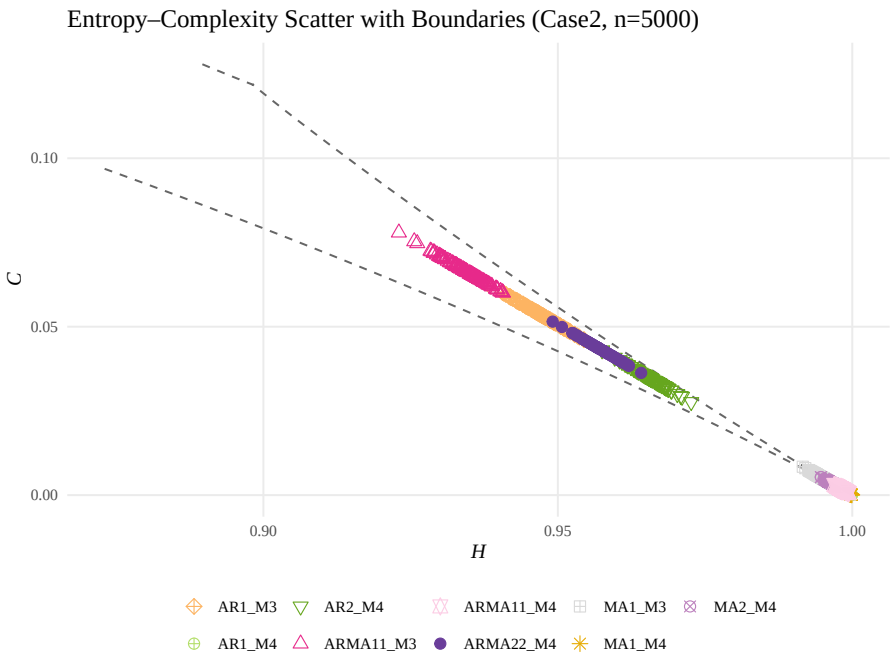
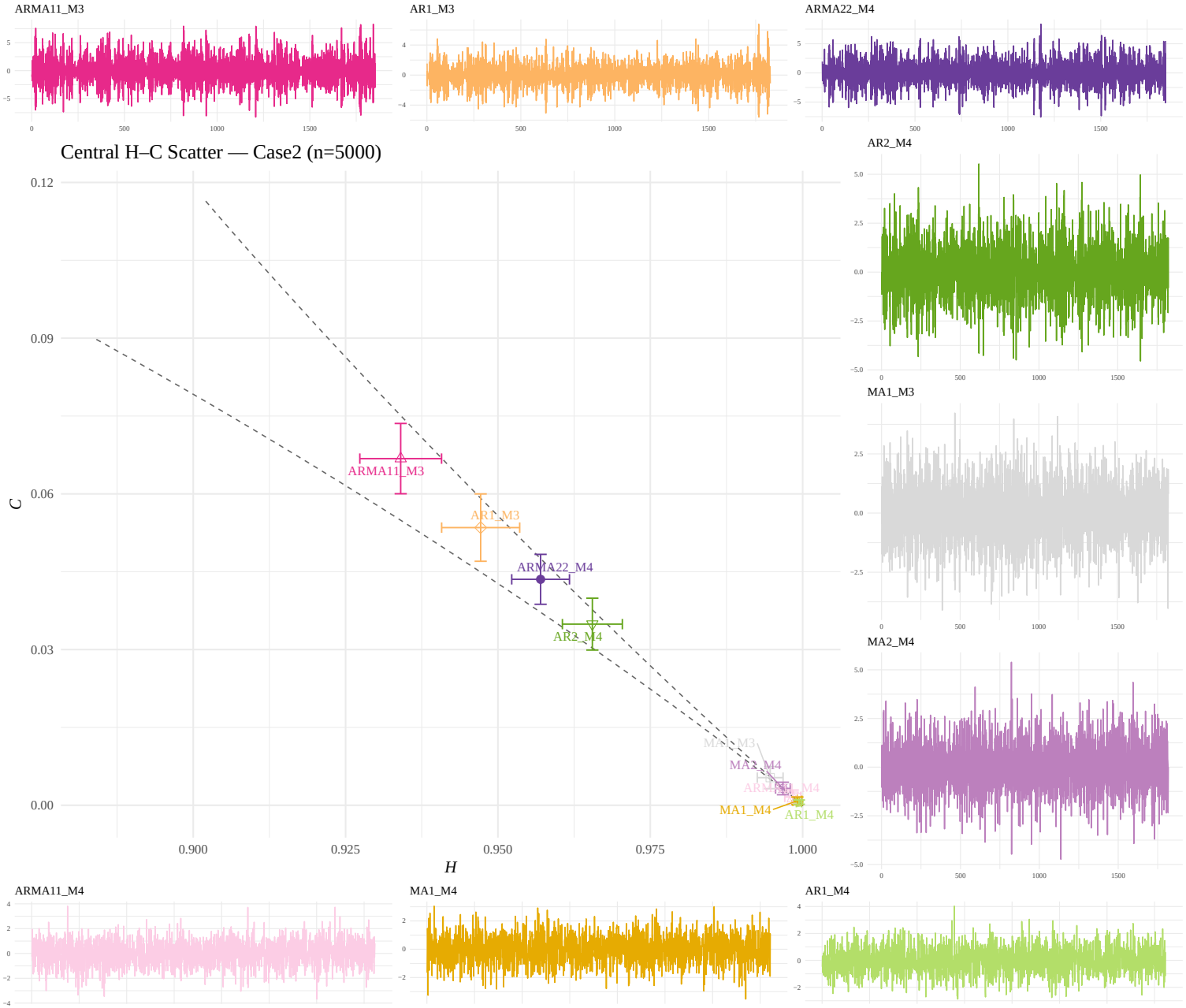


Figure: Entropy–complexity scatter plot ($n = 1000$ and $n = 5000$) for negative model coefficients.

Case 2 — Negative Coefficients cont. . .



Case 3 — mixed-Sign Coefficients

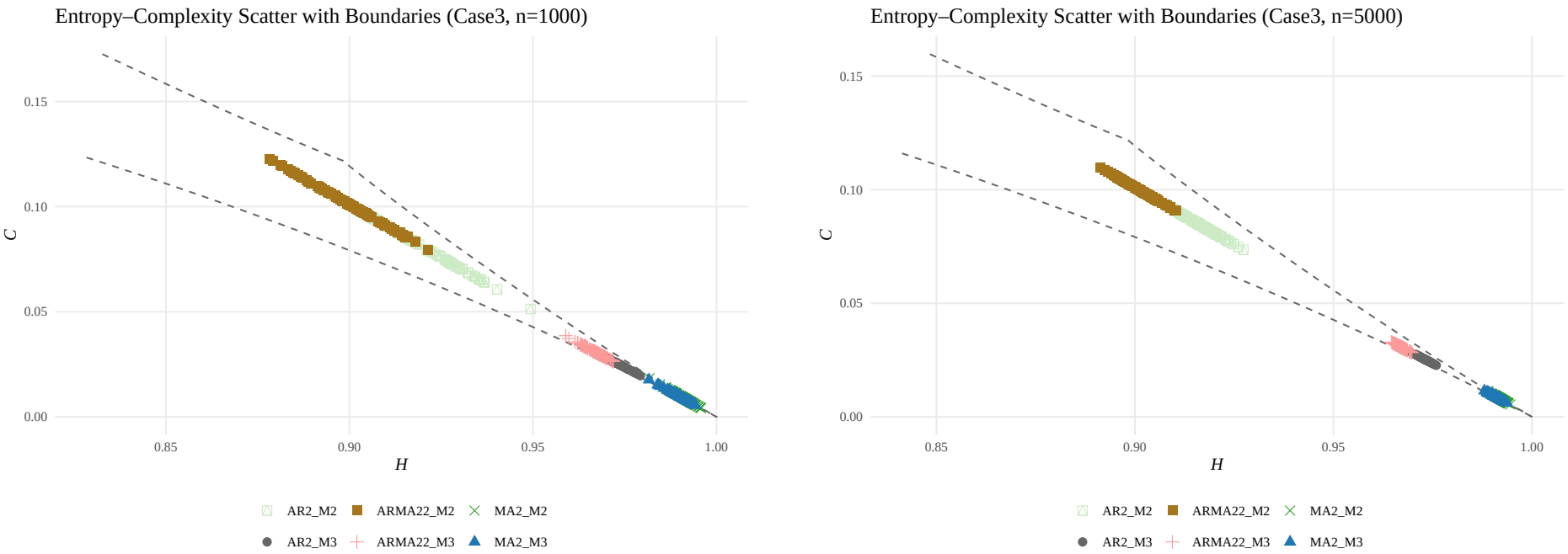
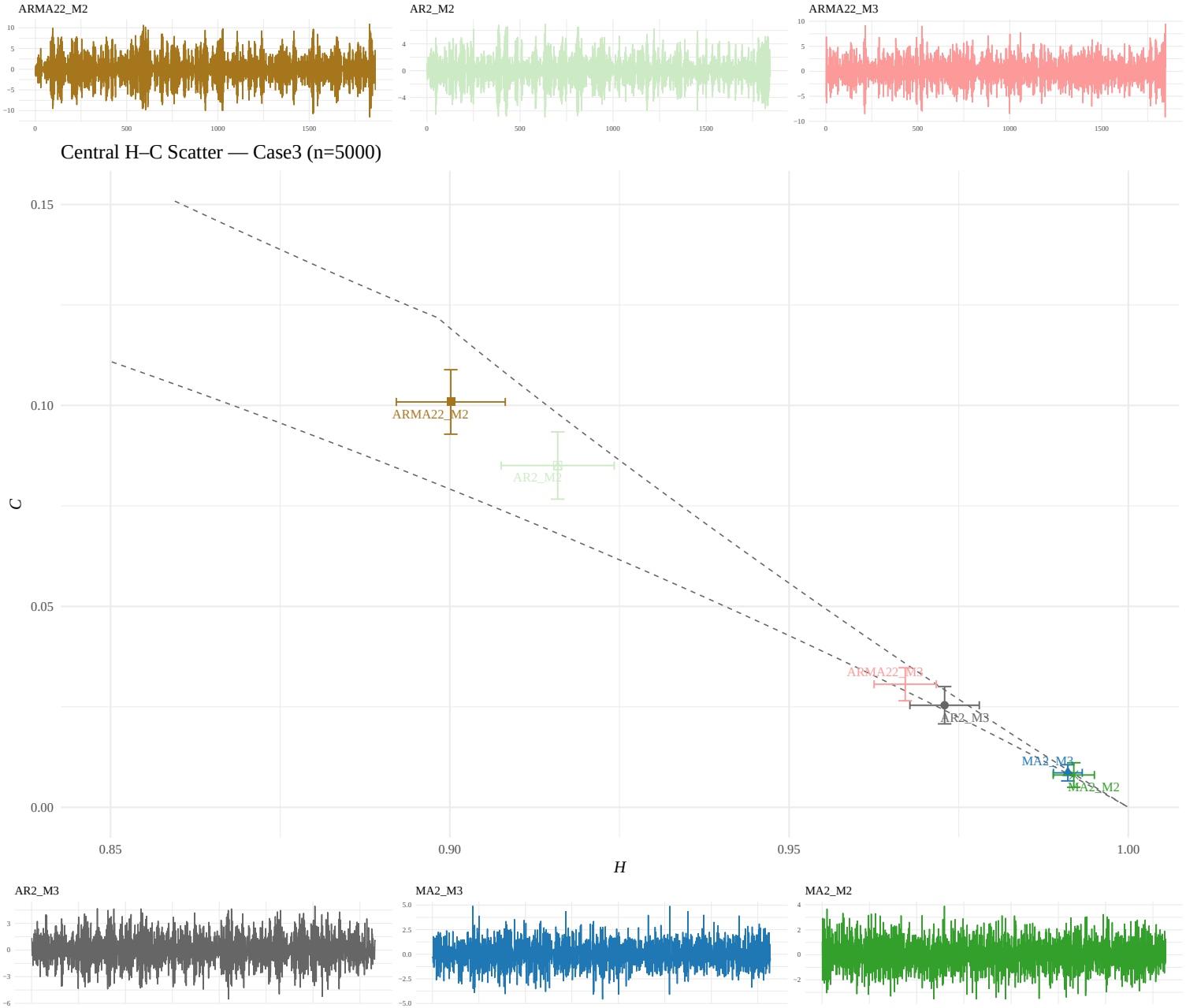
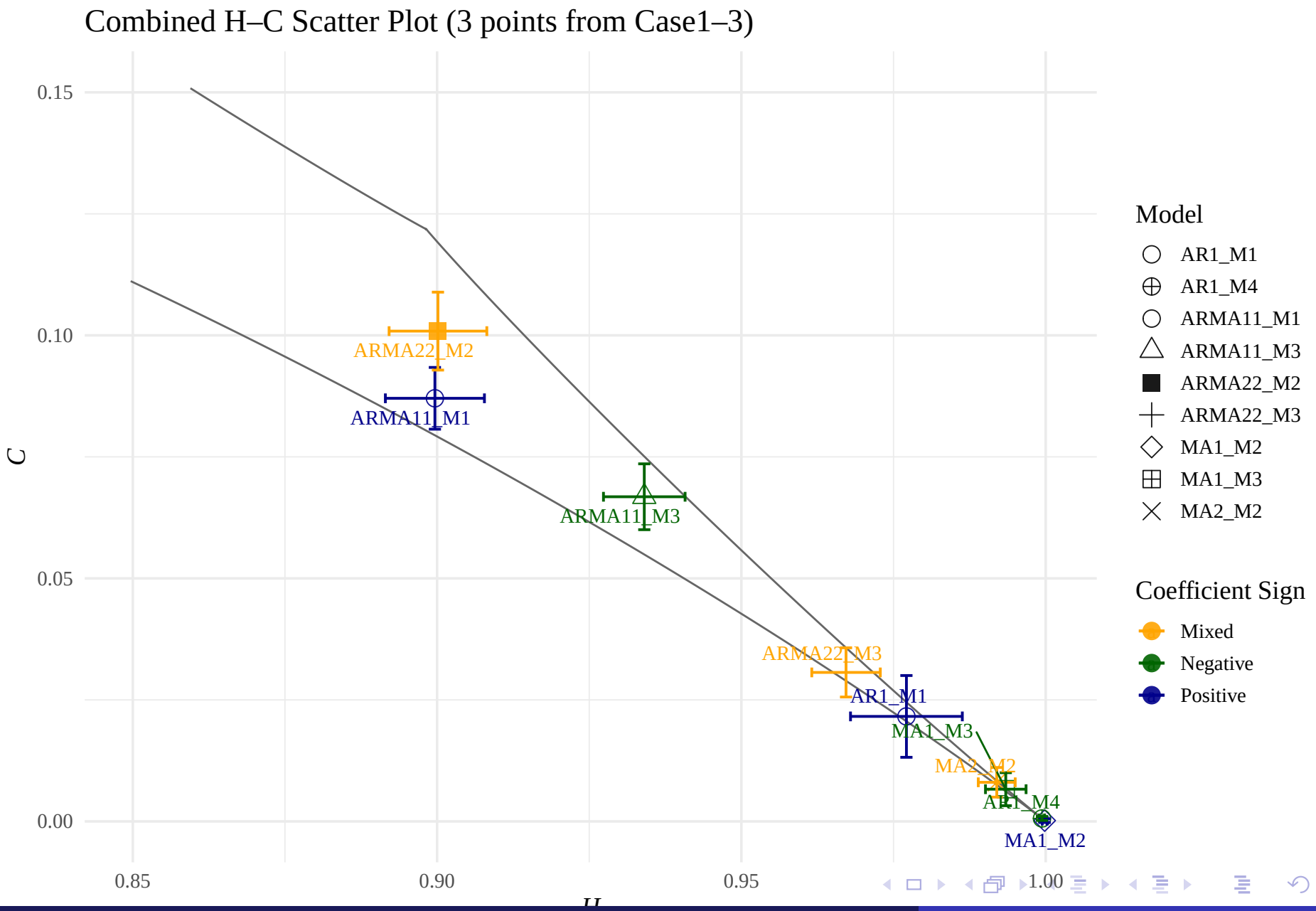


Figure: Entropy–complexity scatter plot ($n = 1000$ and $n = 5000$) for mixed model coefficients.

Case 3 — Mixed-Sign Coefficients cont. . .



Combined Model Visualization



Key Findings

- Ordinal pattern analysis successfully distinguishes AR, MA, and ARMA models.
- Distinct clustering appears across permutation entropy and complexity.
- Central (H^*, C^*) points provide consistent model representation.
- Confidence intervals for permutation entropy and complexity help to distinguish between clusters.
- Larger samples yield tighter clusters and better discrimination.

Thank You!

Contact: rasika.dilhani@vuw.ac.nz

References I

C. Bandt and B. Pompe.
Permutation entropy: A natural complexity measure for time series.
Phys. Rev. Lett., 88:174102, Apr 2002.

E.T.C. Chagas, M. Queiroz-Oliveira, O.A. Rosso, H.S. Ramos, C.G.S. Freitas, and A.C. Frery.
White noise test from ordinal patterns in the entropy–complexity plane.
International Statistical Review, 90(2):374–396, 2022.
cited By 6.